A data project about Boston's gas pipes

Fixing Boston's old gas pipes

Repair, replace, or switch to electric heat? A street-by-street look.

Cathy Kam

The problem

Much of Boston's gas pipe is old cast iron and steel that can leak

34%

of Boston Gas's main pipes are made of leak-prone materials, such as cast iron and unprotected steel.

9,077

gas leaks were still open across Massachusetts at the end of 2024.

Why it matters

Natural gas is mostly methane, a powerful greenhouse gas. Leaks waste gas and can be a safety risk.

Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

What's been done

Since 2015, Massachusetts has paid to replace leaky pipes

$4.75B

spent from 2015 to 2024 under the Gas System Enhancement Program (GSEP).

2,411

miles of leak-prone gas main replaced across the state in that time.

Who pays

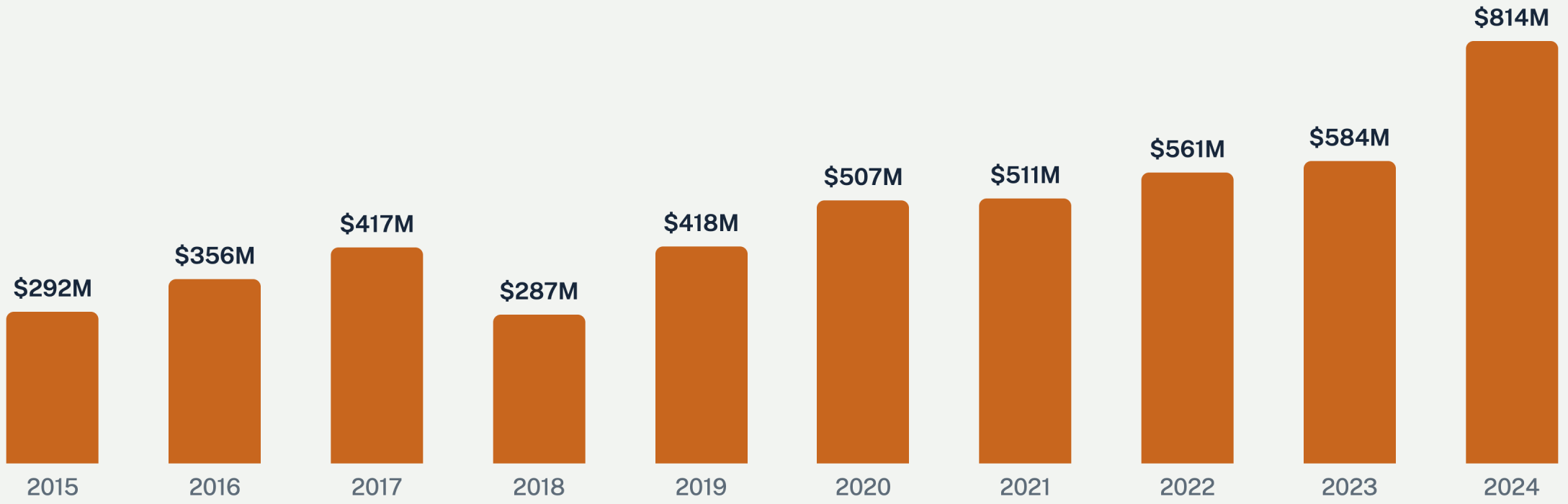
The cost is added to customers' gas bills over many years. It is not paid from taxes.

Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

The cost

Yearly spending nearly tripled

Money spent each year on replacing leak-prone pipe, statewide

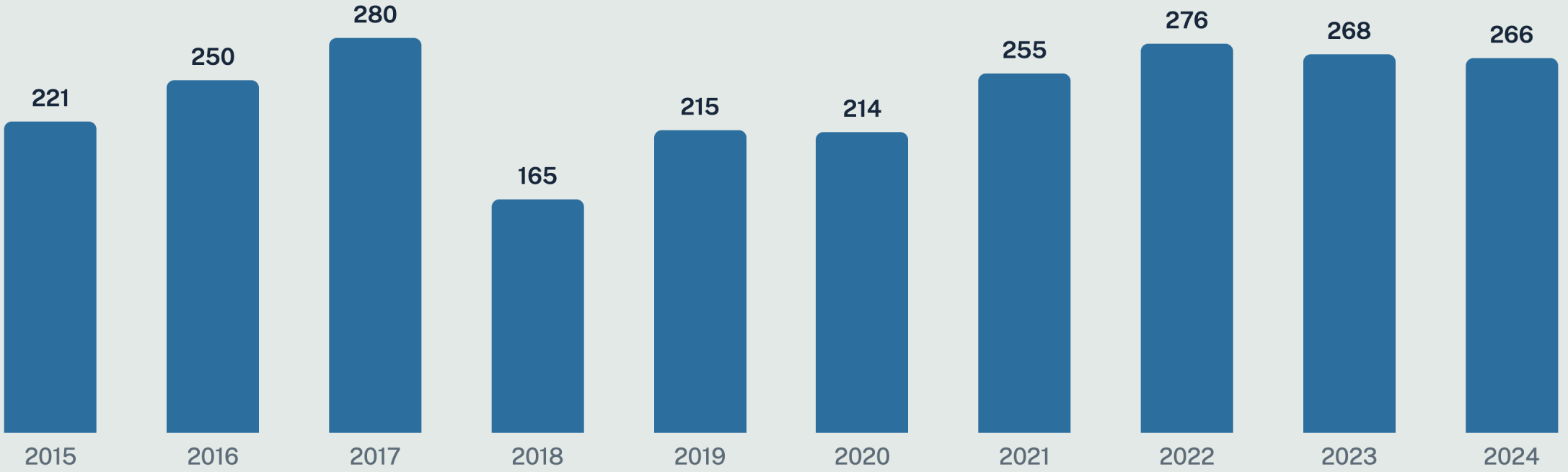


Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025). Includes mains, service lines and related work; not adjusted for inflation.

The progress

But the miles replaced each year barely changed

Miles of leak-prone gas main replaced each year, statewide



Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

Cost per mile

$3.06 million

is what replacing one mile of gas pipe cost in 2024, up from $1.32 million in 2015.

Calculated from DPU-reported spending and miles. Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

Did it help?

Yes: open gas leaks fell by more than half

End of 2014

20,775

open leaks statewide



End of 2024

9,077

open leaks statewide, 56% fewer

Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

The new question

In 2025, regulators told gas companies to look at lower-cost options

Options besides new pipe

Seal the joints of old mains from inside, or switch the homes on a street to electric heat pumps and retire the old pipe.

A real test

Gas companies reviewed 500 possible alternative projects for 2026. None were judged worth pursuing.

What is the lowest-cost way to make each street safe, and how sure can we be?

Sources: DPU GSEP orders (April 30, 2025); DPU data reported by CommonWealth Beacon (2026).

How I answered it

Four steps, all with public data

1

Gather the data

About 17,800 Boston street segments, joined with building records, energy use, Census data and state reports.

2

Map the network

Find dead-end branches that could come off gas without cutting off other streets.

3

Compare three options

Replace, seal or electrify, with owner consent, timing and crew capacity counted.

4

Share the results

An interactive app with a live map, where anyone can change the assumptions.

Finding 1

The tipping point is about 8.6 homes per 100 meters of street

New pipe costs about **$3.06 million** per mile.

A heat pump for one home costs about **$22,000** (an estimate; prices vary).

$3.06 million buys heat pumps for about **139 homes**, which works out to 8.6 homes for every 100 meters of street.

Fewer homes

Switching the homes to heat pumps costs less than new pipe.

More homes

Laying new pipe costs less, because its cost is shared by more homes.

Finding 2

Most of Boston's oldest streets are past the tipping point

351 miles of Boston streets likely sit on old, leak-prone pipe



33 miles: heat pumps cost less, on cost alone

318 miles: pipe work costs less

2,560

homes, roughly, live on streets where heat pumps cost less on cost alone.

18

homes per 100 meters on the typical old street with homes, about twice the tipping point.

82%

of the old streets with homes sit above the tipping point, by mileage.

Finding 3

The gas network and owner consent narrow 33 miles to about half a mile

The 33 miles where heat pumps cost less on cost alone, by position on the network



1.5 miles: dead-end branches that could come off gas

31.4 miles: gas flows past to other streets

1.3 mi

where the plan tries electrification: 39 short branches, 110 homes, 99 owners.

39%

average chance that every owner on a branch agrees. If one declines, the branch needs new pipe.

0.5 mi

of old pipe expected to come off gas once consent is counted.

Network from the City of Boston street map (19,440 segments, 1,875 dead-end branches). Consent: 80% per owner-occupant, 60% per landlord.

Finding 4

Most electrification attempts are in environmental-justice neighborhoods

92%

of the 1.3 miles where the plan tries electrification are in neighborhoods the state identifies as environmental-justice communities.

What that means

Any program there would need income-based financial help, and protections for renters, who don't control their building's heating system.

Environmental-justice areas: MassGIS 2020 Environmental Justice Populations, based on income, race and English-language ability.

Finding 5

The plan: replace, seal, or electrify

259 mi

replace the pipe

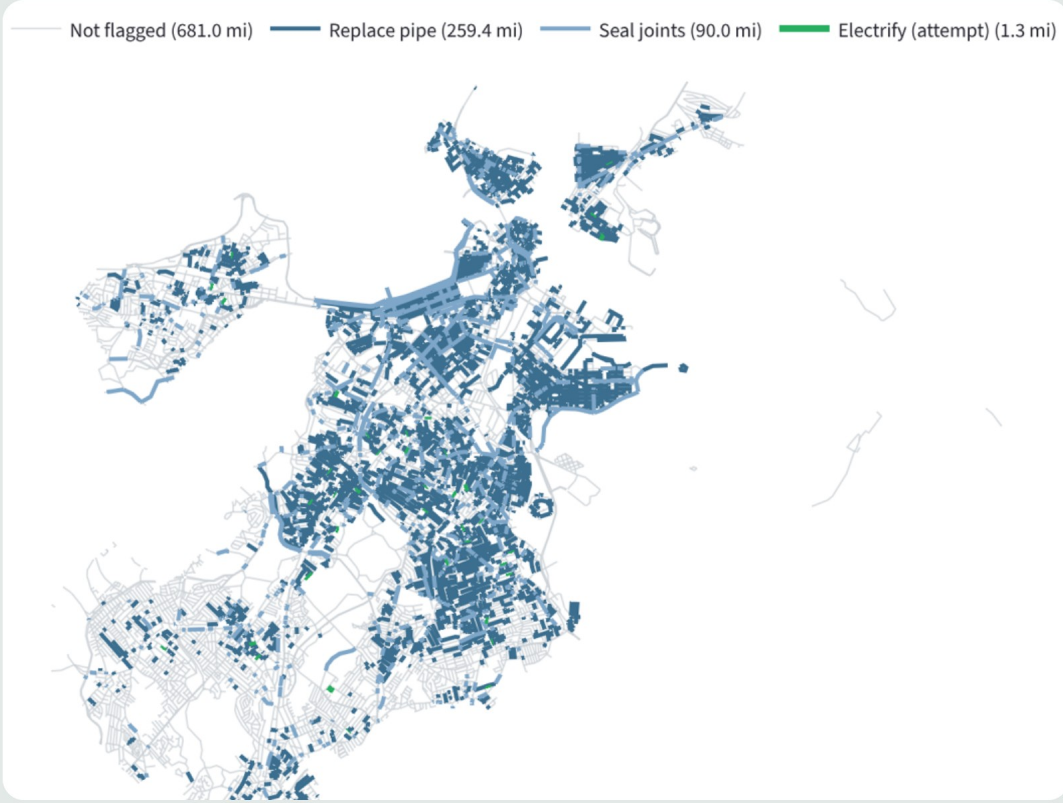
90 mi

seal joints under major roads, about a quarter of the cost

1.3 mi

try electrification on dead-end branches

$1.02B, 5% less than replacing every street ($1.07B), over about **28 years**.



Default settings, capital cost. $50M a year and 17 crew-miles, falling with DPU's spending cap; highest-risk 10% first.

How sure can we be?

Most choices are firm. The close calls are where pilots would help most.

349 mi

where the same option wins anywhere within the cost ranges.

62

dead-end branches (2.2 miles) where pipe work and electrification could swap places.

43 / 19

pipe currently cheaper on 43; electrification on 19. Either could flip.

A pilot on a close-call street, with real costs and real sign-ups shared openly, would settle the question in either direction.

Cost ranges (assumptions): replace ±20%, electrify ±30%, seal ±40%, wider where there is less real-world data.

Honest limits

What this analysis can't tell us yet

Exact pipe locations

Old pipe is estimated from building age, not from utility pipe or leak records.

Local costs

It uses the statewide cost per mile. Boston's streets may cost more or less.

The real network

Dead ends come from the street map; utility maps must confirm the gas mains match.

Costs left out

Panel and grid upgrades, future repairs on new pipe, and monthly bills. The net effect is uncertain.

Takeaways

- Replacing pipe has cut leaks, but the cost per mile has more than doubled.
- Heat pumps cost less than new pipe on about 33 miles; the network and owner consent narrow that to about 0.5 miles.
- Replacing and sealing are lower-cost on most streets: a plan about 5% cheaper than replacing everything, over about 28 years.
- 62 close-call branches are where pilots would tell us the most.

Explore the interactive app:

boston-gas-triage-kljspjcnsvm6y8pc8uv3sp.streamlit.app

Cathy Kam. Built with public data; not an official DPU or utility analysis.